

ONLINE SHOPPERS PURCHASING INTENTION

Presented by
Ramya Balachandran
III BCA – B



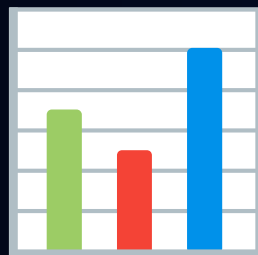
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INTRODUCTION

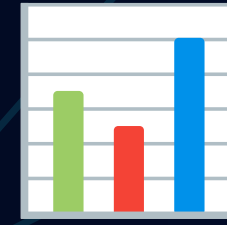
- Online shopping is an important part of e-commerce.
- This project analyses online shopping behaviour using Machine Learning.
- The main goal is to predict whether a customer will make a purchase.
- Revenue is used as the target variable.

DATASET SOURCE & DETAILS



- Source: UCI Machine Learning Repository
- Dataset: Online Shoppers Purchasing Intention
- Dataset ID: 468
- Size: 12,330 sessions
- Target: Revenue
- Purpose: Predict whether a session ends in a purchase.

DATASET DESCRIPTION



- Rows: 12,330
- Columns: 18
- Input Features: 17
- Target Variable: Revenue
- Features: Numerical and categorical data
- Target: Purchase or No Purchase

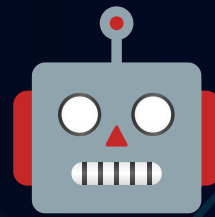
EXPLORATORY DATA ANALYSIS (EDA)

- Dataset Shape – Rows and columns checked.
- Data Types – Numerical and categorical features identified.
- Missing Values – Checked for null values.
- Class Distribution – Purchase and no-purchase classes analysed.
- Feature Analysis – Customer browsing behaviour analysed.

DATA PREPROCESSING

- Label Encoding – Converted categorical values into numbers.
- One-Hot Encoding – Converted categorical features into separate columns.
- SMOTE – Balanced the purchase and no-purchase classes.
- SelectKBest – Selected the top 10 important features.
- StandardScaler – Scaled the selected features.
- Train-Test Split – Split the data into 80% training and 20% testing.

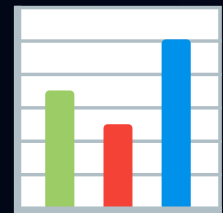
MACHINE LEARNING ALGORITHMS



- Logistic Regression
- Decision Tree
- Random Forest
- AdaBoost
- Gradient Boosting

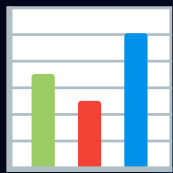
These 5 classification algorithms were used to predict whether the customer will make a purchase or not.

EVALUATION METRICS



- Accuracy – Measures overall correct predictions.
- Precision – Measures how many predicted purchases were actually correct.
- Recall – Measures how many actual purchases were correctly identified.
- F1-Score – Combines Precision and Recall.

MODEL EVALUATION & COMPARISON



Evaluation Metrics:

Accuracy

Precision

Recall

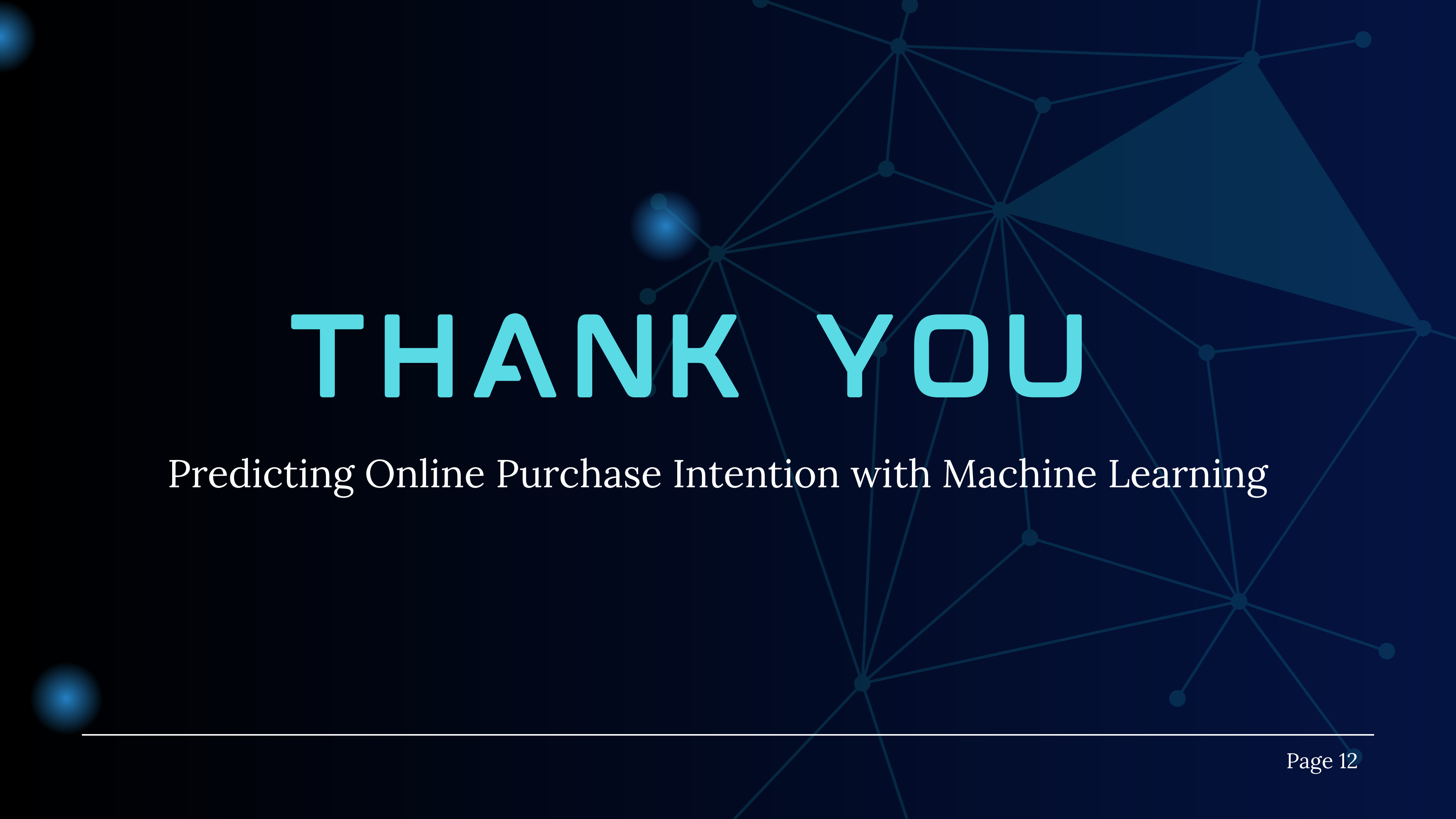
F1-Score

Best Model: Random
Forest

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.68	0.65	0.76	0.7
Decision Tree	0.8	0.79	0.8	0.8
Random Forest	0.87	0.88	0.86	0.87
AdaBoost	0.75	0.72	0.81	0.76
Gradient Boosting	0.83	0.84	0.8	0.82

CONCLUSION

- Machine Learning was used to predict online purchases.
- Random Forest gave the best result.
- It achieved 87% accuracy.
- The project helped analyse customer shopping behaviour.



THANK YOU

Predicting Online Purchase Intention with Machine Learning